Exponential Information Gathering (EIG): Crash Consensus

Curated from Arpit Bhayani's Distributed Systems Architecture Lecture Series • Reference: Lynch et al.

EXECUTIVE ARCHITECTURAL ABSTRACT

The **Exponential Information Gathering (EIG) Algorithm** solves distributed consensus in synchronous networks subject to up to *f* crash-stop failures in exactly **f + 1 rounds**. Unlike flat-set protocols (such as FloodSet), EIG structures knowledge into an explicit **path-labeled attribution tree** of depth f + 1. By recording the complete transmission history ("who reported what to whom"), surviving nodes reconstruct identical global views despite mid-broadcast crash cascades.

1. THE DISTRIBUTED CRASH CONSENSUS MODEL

Consider a synchronous network of *n* processes P = {p1, p2, ..., pn} connected by reliable, delay-bounded FIFO channels. At time t = 0, each process pi holds a private initial input vi in V. The cluster must satisfy three canonical consensus properties:

- **Agreement:** Every non-faulty process must output the identical decision value v* in V.
- **Validity:** If all processes begin with initial value v, all non-faulty processes must decide v. If inputs differ, the decision must belong to V.
- **Termination:** Every non-faulty process must decide and halt within a strictly bounded number of rounds.

The Mid-Broadcast Crash Threat

A node broadcasting to n-1 peers may crash halfway through transmission. A subset S of peers receives the value, while the remaining peers receive nothing (bot). In flat-set models, this creates asymmetric knowledge divergence. EIG preserves the transmission path so surviving peers can detect missing data and reconcile divergence.

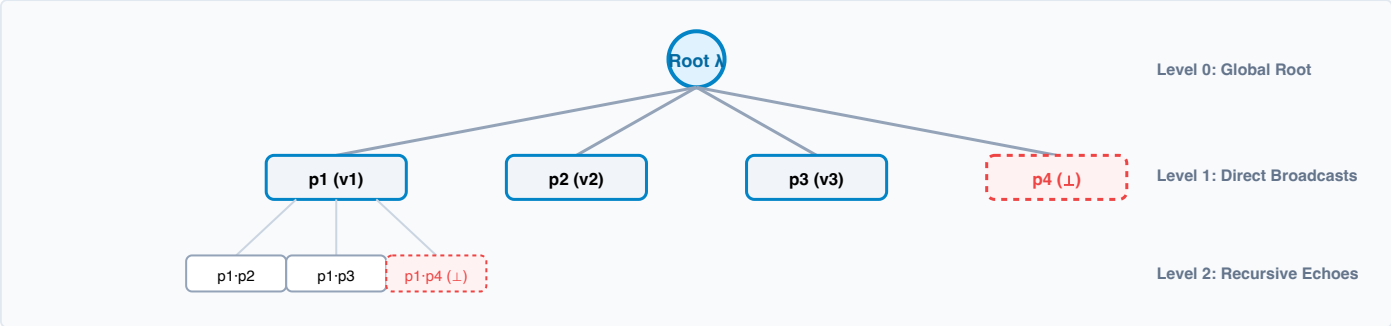
The f + 1 Round Lower Bound

Consensus under f crash failures cannot be solved in fewer than f + 1 synchronous rounds. By the **Pigeonhole Principle**, distributing at most f crashes over f + 1 rounds guarantees that at least one round experiences *zero crashes* (the Clean Round).

2. FORMAL DEFINITION OF THE EIG TREE STRUCTURE

Each process pi maintains a private local tree Ti characterized by the following properties:

- **Depth & Height:** Exactly f + 1 levels. Root is at level 0; leaves reside at level f + 1.
- **Node Path Labels:** A node at depth k is identified by a sequence of distinct process identifiers α = j1 j2 ... jk with 1 ≤ k ≤ f + 1.
- **Attribution Semantics:** The label α = j1 j2 ... jk signifies: "jk reported that jk-1 reported that ... that j1 initially started with value vj1."
- **Stored Values:** For each node α, val(α) in V union {bot}. Here bot explicitly denotes an unreceived message (node crashed).



3. PROTOCOL EXECUTION ACROSS F + 1 ROUNDS

The EIG crash consensus protocol proceeds across strictly synchronized rounds r = 1, 2, ..., f + 1:

Round 1: Initial State Exchange

1. Every process pi broadcasts its initial private value vi to all peers.
2. For every peer pj, pi sets val(j) = vj if a message arrives before round timeout; otherwise val(j) = bot.
3. Level 1 of Ti is fully populated with n nodes.

Round r (2 ≤ r ≤ f + 1): Path Echoing

1. Process pi broadcasts all pairs (α, val(α)) from **level r - 1** of its tree.
2. Upon receiving message from peer pj, for each prefix α not containing j, pi sets: `val(α · j) = incoming_val`.
3. If pj sent nothing, set `val(α · j) = bot`.

4. DETERMINISTIC BOTTOM-UP TREE REDUCTION RULE

Upon completing round f + 1, all surviving processes evaluate their local EIG trees from the leaves upward to derive unanimous consensus:

Mathematical Reduction Operator: resolve(node)

1. Base Case (Leaves at depth f + 1):

For any leaf node α at level f + 1:
`newval(α) = val(α) if val(α) ≠ bot else v_default`

2. Inductive Reduction (Levels f down to 0):

For any internal node α at depth k ≤ f, consider the set of resolved child values:
`Children(α) = { newval(α · j) | j not in path(α) }`

- If all values in `Children(α)` are identical to some unique value `v`, then `newval(α) = v`.
- Otherwise (conflict or divergence due to crashes), apply default tie-breaker: `newval(α) = min(Children(α))`.

3. Terminal Output: The final cluster decision is the resolved root value: `v* = newval(λ)`.

5. REFERENCE PYTHON IMPLEMENTATION (SELF-CONTAINED)

```
class EIGCrashNode:
    def __init__(self, node_id, n, f, initial_val):
        self.id, self.n, self.f = node_id, n, f
        self.val = {}      # Raw tree values: path tuple -> int or None
        self.newval = {}   # Reduced values: path tuple -> int
        self.initial_val = initial_val
        self.val[(self.id,)] = initial_val

    def round_r_payload(self, r):
        # Transmit all nodes at level r - 1
        return {p: v for p, v in self.val.items() if len(p) == r - 1}

    def process_incoming(self, r, incoming_msgs):
        for peer_id, payload in incoming_msgs.items():
            for p, v in payload.items():
                if len(p) == r - 1 and peer_id not in p:
                    self.val[p + (peer_id,)] = v

    def reduce_tree(self):
        # 1. Base case: leaves at f + 1
        for p, v in self.val.items():
            if len(p) == self.f + 1:
                self.newval[p] = v if v is not None else 0
        # 2. Bottom-up reduction
        for level in range(self.f, -1, -1):
            paths = [p for p in self.val if len(p) == level] if level > 0 else [()]
            for p in paths:
                c_vals = [self.newval[p + (j,)] for j in range(1, self.n+1) if (p + (j,)) in self.newval]
                self.newval[p] = c_vals[0] if (c_vals and all(x == c_vals[0] for x in c_vals)) else (min(c_vals) if c_vals else 0)
        return self.newval[()]
```

6. CORRECTNESS PROOF & THE CLEAN-ROUND INVARIANT

The mathematical guarantee of agreement across all non-faulty processes rests upon the Pigeonhole Principle:

Theorem: Consensus Agreement Under f Crashes

In any synchronous execution with at most f crash failures, after f + 1 rounds all non-faulty processes pi, pj satisfy `newval_i(λ) = newval_j(λ)`.

Proof Steps:

1. **Pigeonhole Partitioning:** Since at most f processes crash across f + 1 execution rounds, there exists at least one round k in {1, 2, ..., f + 1} with *zero crashes* (the Clean Round).
2. **Universal State Synchronization:** In clean round k, every process alive transmits its entire level k - 1 tree to all surviving peers without message loss. Hence, for every path α of length k, all non-faulty nodes record identical values: `val_i(α) = val_j(α)`.
3. **Downward Propagation:** Subsequent transmissions in rounds k+1 to f+1 are deterministic copies of level k. Even if nodes die later, they cannot corrupt the synchronized values at level k.
4. **Bottom-Up Induction:** Since all children of node α at level k evaluate to identical values across all surviving nodes, induction proves `newval_i(α) = newval_j(α)` up to root λ.

7. ALGORITHMIC COMPARISON: EIG VS. FLOODSET VS. PAXOS

Dimension	EIG Algorithm (Crash)	FloodSet Algorithm	Paxos / Raft (Async Quorums)
Failure Model	Crash-stop (up to f < n)	Crash-stop (up to f < n)	Crash-recovery (f < n/2)
Rounds to Decide	f + 1 rounds (Optimal bound)	f + 1 rounds (Optimal bound)	1–2 RTTs typical (Unbounded worst-case)
State / Message Overhead	O(n^{f+1}) nodes (Exponential)	O(|V|) values (Minimal / Flat)	O(Log Entries) (Linear)
Provenance Tracking	Complete tree path history	None (Flat value sets)	Term numbers and ballot ranks
Byzantine Applicability	**Yes** (Foundational for 3f+1 BA)	**No** (Poisoned by single lie)	Requires PBFT / Tendermint

8. SYSTEMS ENGINEERING TAKEAWAYS

1. **Theoretical Value vs. Operational Reality:** EIG demonstrates that maintaining complete message provenance guarantees consensus. However, its O(n^{f+1}) state makes it an intellectual tool rather than a database protocol. Practical crash-resilient systems use FloodSet (for synchronous lockstep) or Paxos (for asynchronous quorums).
2. **The Irreducible Cost of Crash Safety:** Whether using lightweight FloodSet or attribution-rich EIG, tolerating f crash failures in a synchronous network mathematically requires f + 1 rounds of communication.
3. **The Gateway to Byzantine Agreement:** In crash failures, nodes only omit messages (bot). In Byzantine models, nodes send contradictory values. The EIG tree is the exact vehicle used by Pease, Shostak, and Lamport to solve Byzantine Agreement via recursive majority voting.